Data-Driven Quantum-System Prediction with WPM

Observable forecasts across wave, lattice, spin, light–matter and open-system families

Thy Nguyen

29 October 2026
Mathematical methods manuscript, revised edition
DOI: 10.5281/zenodo.23783870

Abstract

This paper develops an observation-limited forecasting procedure for the quantum families considered in the WPM research program. It separates quantum probabilities from uncertainty about their future evolution, defines target-conditioned resolution landscapes, and presents family-specific observation contracts for wave packets, interacting lattices, spin chains, light–matter systems and dissipative dynamics. Hamiltonians and master equations identify the physical meaning of examples; the forecasting layer is trained on histories rather than supplied the true generator. Exact examples, constraint-preserving constructions and identifiability arguments specify how to apply the method without interpreting pockets as physical quantum wells or forecasts as deterministic measurement outcomes.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Observation-limited forecasting and quantum probabilities

Let $\mathcal{F}_t$ contain measurements, estimated observables, known settings and context available at issuance. A quantum state is $\rho_t \succeq 0$ with $\operatorname{tr}\rho_t = 1$. For an observable O, its expectation is $z_O(t) = \operatorname{tr}(\rho_t O)$. A POVM outcome a has probability $p_a(t) = \operatorname{tr}(\rho_t E_a)$, where $E_a \succeq 0$ and $\sum_a E_a = I$. These physical definitions do not imply that the predictor has access to the complete state.

The forecasting task is to estimate a future observable, occupation probability, density profile, structural event or uncertainty region from the observed prefix. Write

$$\widehat{Y}^T_{t,P} = f_{\theta,T}(D_{\le t}; W, m, P, S_t, C_t). \tag{1.1}$$

The fitted θ comes from permitted histories and matured labels. The physical Hamiltonian used to generate a reference trajectory is not an inference input unless a separately labeled physics-informed variant is chosen. A known drive schedule is usable when available at issuance; an unknown adaptive drive is not a known future input.

There are two probability levels. $B_R(t) = \mathrm{tr}(\rho_t \Pi_R)$ is the Born probability of a measurement in region R at time t. By contrast, $\mathbb{P}(B_R(t+P) > b \mid \mathcal{F}_t)$ describes uncertainty about a future probability functional. An event involving the actual sequence of measurements needs an explicit measurement protocol, including disturbance; it is not automatically defined by an unmeasured state trajectory.

2 Resolution landscapes and target-specific pockets

Fix physical horizon q or sample horizon $P = q/\Delta t$ under a declared sampling convention. W controls historical extent and m the internal sampling or representation resolution. For each target T use only matured issued losses:

$$\widehat{L}_t(W, m; T, P) = \frac{\sum_{s:\, s+P \le t} w_s \ell_T(Y^T_{s,P}, \widehat{Y}^T_{s,P,W,m})}{\sum_s w_s}. \tag{2.1}$$

The denominator runs over the same matured origins and must be positive. Missing risk is unknown. A pocket is a connected low-risk region of this surface under a declared adjacency and threshold. It lies in representation space, not necessarily physical position.

Retain each pocket's complete flow cross-sections, position, existence, support, signed errors and identity history. Separate horizon, observable, system and regime layers. Fixed chooses an eligible reference lens; Multi combines same-target alternatives; Auto updates a specified weight policy from causal evidence. None is guaranteed best for every family.

Proposition 2.1 (Target-conditioned optimality)**.** *An optimal lens for mean position need not optimize width, region occupation or a boundary event.*

Proof. The losses are different functionals of the forecast. Two probability distributions can share mean and variance but assign different masses to R. A predictor correct on those moments can therefore be wrong on occupation, while another predictor reverses that ordering. Without additional distributional restrictions the minimizers need not coincide. □

For a concrete moment ambiguity, the distributions assigning equal mass to $\{-1, 1\}$ and masses $(1/4, 1/2, 1/4)$ to $\{-\sqrt{2}, 0, \sqrt{2}\}$ both have mean 0 and variance 1; their mass in $[-1/2, 1/2]$ is 0 and 1/2. A Gaussian conversion from mean and width silently excludes such alternatives.

3 Observation and target catalogue

Family	Possible observed channels	Example future target
Free packet; harmonic oscillator	Density snapshots, centroid, width, current if available	Density, position, width or region occupation
Barrier scattering	Incident/transmitted density and measured flux	Transmission occupation or crossing event
Double well; multi-well	Region populations and density geometry	Population transfer, width or occupation excursion
Coherent interference	Resolved fringes and phase-sensitive channels when measured	Fringe-region density or contrast
Gross–Pitaevskii	Condensate density and optional measured phase/current	Center, width or specified tail mass
Lindblad open system	Populations, coherence, purity estimates	Decoherence, purity or state expectation
Transverse-field Ising	Magnetization, correlations, domain-wall proxies	Magnetization excursion or structural transition
Anderson lattice	Site probabilities, width, participation and disorder settings	Localization episode
Bose–Hubbard; hard-core/spinless	Density, fluctuations, correlations and potential profile	Occupation redistribution or density excursion
Jaynes–Cummings; full Rabi	Atomic excitation, photon statistics and coherence	Excitation or revival excursion
Floquet-driven system	Phase-tagged observables and drive settings	Stroboscopic or within-cycle event
Quantum kicked rotor	Momentum probabilities, variance and kick settings	Normalized transport excursion
Non-Markovian open system	State-pair distinguishability or qualified amplitude channels	Future backflow functional

This catalogue covers the families used in the research lineage. It is not a claim that every observable was measured experimentally or that every proposed channel was used in an archived implementation. Density snapshots and state estimates require actual ensembles, repeated preparations or an appropriate measurement scheme; one cannot obtain a full wavefunction from a single position measurement.

4 Free packets and harmonic motion

A familiar one-dimensional physical model is $i\hbar\partial_t\psi = [-\hbar^2\partial_x^2/(2M) + V(x)]\psi$. It supplies a reference example, not an equation fed to the data-driven predictor. A minimum-uncertainty Gaussian with initial position SD s_0, center x_0 and mean momentum p_0 under free evolution has

$$\mu_x(t) = x_0 + p_0 t/M, \quad s_x^2(t) = s_0^2 + \frac{\hbar^2 t^2}{4M^2 s_0^2}. \tag{4.1}$$

For a quadratic oscillator $V = M\omega^2 x^2/2$, the mean obeys $\mu_x(t) = x_0\cos\omega t + [p_0/(M\omega)]\sin\omega t$. A coherent-state width is constant; a squeezed-state width need not be. Thus the observable target matters even within a single family.

Worked example 4.1 (A physical probability target)**.** For a Gaussian density with predicted mean 1 and SD 1/2, the occupation of $[0, 2]$ is $\Phi(2) - \Phi(-2)$. This formula is valid only under

the Gaussian shape model. A data-driven density forecast can instead integrate its non-Gaussian predicted density over the interval.

The required inference inputs are observed histories with sufficient sampling of motion and spreading. Supplying ω, p_0 and integrating the exact oscillator law defines a different, equation-informed comparator. Comparing its state error to an event forecaster's Brier score is not meaningful.

5 Barrier scattering, double wells and multiple wells

For barrier scattering, specify whether transmission means instantaneous mass beyond a boundary, asymptotic transmitted flux or a finite-horizon arrival event. These differ when reflected and transmitted packets coexist. If $\rho = |\psi|^2$, region occupation is $B_R = \int_R \rho\, dx$. Boundary flux uses the current $j = (\hbar/M)\,\mathrm{Im}(\psi^* \partial_x \psi)$ and requires more than density at one time.

A double-well reference potential is $V(x) = a(x^2 - b^2)^2 + cx$. Choose left/right regions explicitly; the tilt c changes the physical symmetry. A two-state approximation with Hamiltonian $H = -J\sigma_x$ and initially left-localized state yields $B_R(t) = \sin^2(Jt/\hbar)$. Derive this by $e^{iJt\sigma_x/\hbar} = \cos(Jt/\hbar)I + i\sin(Jt/\hbar)\sigma_x$ and take the right-state amplitude squared.

Worked example 5.1 (Transfer is a target, not a pocket count)**.** At $Jt/\hbar = \pi/6$, the two-state model has right occupation $1/4$. An excursion target $\mathbf{1}\{\max_{h \le P} B_R(t+h) > 0.4\}$ requires the future path, not only the current $1/4$. A W,m risk basin for this target is not one of the two physical wells.

For multiple wells, retain the population vector $(B_1, \ldots, B_K)$ with its sum constraint and physical region labels. Width may be predictable while the identity of the occupied region is uncertain. Averaging incompatible site labels into a fictitious midpoint is not a physical occupation forecast.

6 Coherent interference and the missing-phase limitation

For $\psi = \psi_1 + e^{i\varphi}\psi_2$,

$$|\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\,\mathrm{Re}(e^{i\varphi}\psi_2\psi_1^*). \tag{6.1}$$

The cross term is a quantum amplitude identity. Signed WPM probe interference has a similar quadratic expansion, but the two objects are not interchangeable. An empirical signed forecast error is not a quantum amplitude or a measured phase.

Theorem 6.1 (One density snapshot need not determine future density)**.** *There exist states with equal present density and different future densities under the same known free-particle Hamiltonian.*

Proof. Take the same real Gaussian envelope and multiply it by $e^{ipx/\hbar}$ or $e^{-ipx/\hbar}$, with $p \neq 0$. Present absolute squares agree. Their future means move with velocities p/M and $-p/M$, so future densities differ. □

Temporal density histories can help infer motion, and measured current or phase-sensitive channels can add information. A predictor must still preserve the uncertainty in unobserved phase or unrepresented states. Fine fringe structure requires adequate spatial sampling; numerical interpolation cannot recover arbitrary unmeasured fringes.

7 Gross–Pitaevskii condensate dynamics

A reference mean-field equation is

$$i\hbar\partial_t\psi = \left[-\frac{\hbar^2}{2M}\partial_x^2 + V(x,t) + g|\psi|^2\right]\psi. \tag{7.1}$$

With real V,g and suitable vanishing or periodic boundary flux, the norm $\int |\psi|^2 dx$ is conserved. Multiply the equation by ψ^*, subtract its conjugate and integrate: the real potential and nonlinear terms cancel, leaving a boundary-current term. This proof gives a constraint for a density forecast, not a guarantee that its temporal model is correct.

Define a tail region before prediction. A fixed spatial region, a moving region attached to the current mean and a region attached to the future mean are different targets with different issuance and label requirements. Density normalization and positivity should be enforced; they do not determine the correct tail mass. A context model may use current shape, population allocation and matured error history, while future forcing must be known by the task or forecast from the prefix.

8 Lindblad state and coherence forecasts

A Markovian reference master equation is

$$\dot\rho = -\frac{i}{\hbar}[H,\rho] + \sum_\ell \gamma_\ell \left(L_\ell \rho L_\ell^\dagger - \tfrac{1}{2}\{L_\ell^\dagger L_\ell, \rho\}\right). \tag{8.1}$$

With appropriate nonnegative rates and finite-dimensional conditions it generates physical density matrices. Measured channels can include excitation, Bloch components, purity $\operatorname{tr}\rho^2$ and coherence. A single population history need not identify coherence or environmental memory.

For a qubit write $\rho = (I + \boldsymbol{r}\cdot\boldsymbol{\sigma})/2$. Physicality is equivalent to $\|\boldsymbol{r}\| \le 1$, and purity is $(1+\|\boldsymbol{r}\|^2)/2$. A vector predictor returning $\|\widehat{\boldsymbol{r}}\| > 1$ is unphysical. Projection onto the unit ball is a constraint repair, not proof of predictive accuracy. The observation model must include finite-shot estimation error when those channels are measured rather than supplied as exact expectation values. Standard composite-system and master-equation conventions are summarized in [6].

9 Ising and Anderson lattice families

For a transverse-field Ising reference,

$$H = -J\sum_i \sigma_i^z\sigma_{i+1}^z - h(t)\sum_i \sigma_i^x. \tag{9.1}$$

A possible target is whether $|M_z|$ falls below a declared boundary during the horizon. Finite-size coordinates such as $(h/J-1)N^{1/\nu}$ and $M_z N^{\beta/\nu}$ are physically motivated only with an appropriate scaling regime; exponents do not certify finite-chain forecasting invariance. Correlations and domain-wall descriptors can supplement, not replace, target-specific histories.

For the Anderson chain,

$$i\hbar\dot\psi_n = -J(\psi_{n+1}+\psi_{n-1}) + \epsilon_n\psi_n. \tag{9.2}$$

Let $p_n = |\psi_n|^2$, $\sum_n p_n = 1$, $\mu = \sum_n x_n p_n$, $s^2 = \sum_n (x_n-\mu)^2 p_n$ and $\mathrm{IPR} = \sum_n p_n^2$. A localization event can require high IPR and small width simultaneously. IPR measures concentration but

loses location: a localized packet at either edge can have the same value. Retain density geometry and disorder location rather than treating IPR as a complete state [8].

10 Bose–Hubbard and hard-core/spinless systems

The Bose–Hubbard reference has hopping, onsite repulsion and potential terms,

$$H_{\mathrm{BH}} = -J\sum_i (a_i^\dagger a_{i+1} + \mathrm{h.c.}) + \frac{U}{2}\sum_i n_i(n_i-1) + \sum_i \epsilon_i n_i. \tag{10.1}$$

A hard-core/spinless reference may instead contain nearest-neighbor interaction $V\sum_i n_i n_{i+1}$. Changing statistics, boundary conditions or local occupancy changes the physical model. Density profiles can share a normalized geometric representation without making all correlations or response laws identical.

Observe density, occupation fluctuations, spatial potential and any independently available correlations. Normalize density mass and domain coordinates with explicit conventions. A common target might be a density-redistribution event or right-side population excursion, not the equality of the two Hamiltonians. Optional supplied coupling settings are context; if unavailable, estimate them with uncertainty or omit them. Calling a forecast data driven does not require pretending that known instrument settings are unknown.

11 Jaynes–Cummings and full quantum Rabi systems

The rotating-wave reference is

$$H_{\mathrm{JC}} = \hbar\omega_c a^\dagger a + \frac{\hbar\omega_q}{2}\sigma_z + \hbar g(a\sigma_+ + a^\dagger\sigma_-). \tag{11.1}$$

For a resonant initial state $|e,n\rangle$, excitation in its invariant two-state block is $P_e(t) = \cos^2(g\sqrt{n+1}\,t)$. The full Rabi model replaces the coupling by $\hbar g\sigma_x(a+a^\dagger)$ and retains counter-rotating terms. The JC reduction is not uniformly valid at ultrastrong coupling.

Useful observation channels include excitation, photon-number moments, atom–field covariance, reduced-qubit purity and measured parity. A natural coupling clock $g\sqrt{\bar{n}+1}$ is a context feature only if its ingredients are available. Excitation revival, excursion from the issuance value and a threshold crossing are different events. For a population of photon numbers, a sum of several Rabi frequencies can produce collapse/revival structure; one fitted sinusoid need not extrapolate it.

12 Floquet and kicked-rotor prediction

For a periodic Hamiltonian $H(t+\mathcal{T}) = H(t)$, stroboscopic observations can be modeled using the one-period propagator. If a predictor omits drive phase while mixing within-period observations, it may average physically different states. Choose stroboscopic targets or include a causally known phase. Floquet descriptions of dissipative driven systems also require appropriate open-system assumptions [7].

A kicked rotor has the reference form $H = p^2/(2I) + K\cos\theta\sum_j \delta(t-j\mathcal{T})$ under a declared convention. A dimensionless transport target may use

$$R_p(t,h) = \frac{\mathrm{Var}[p(t+h)] - \mathrm{Var}[p(t)]}{D_{\mathrm{ref}}(K)h}, \qquad D_{\mathrm{ref}}(K) \approx K^2/2. \tag{12.1}$$

The reference diffusion scale is a normalization convention with a restricted regime of usefulness, not a universal quantum diffusion law. Kick strength, effective Planck scale and resonance proximity can change the dynamics. Equal absolute momentum widths across K need not mean equal transport. Event boundaries must transform with the target semantics.

13 Non-Markovian amplitudes and future backflow

One structured-reservoir reference amplitude is

$$G(t) = e^{-(\lambda - i\Delta)t/2}\left[\cosh(dt/2) + \frac{\lambda - i\Delta}{d}\sinh(dt/2)\right], \tag{13.1}$$

$$d^2 = (\lambda - i\Delta)^2 - 2\gamma_0\lambda, \qquad P_e(t) = |G(t)|^2. \tag{13.2}$$

At $d = 0$ interpret the ratio by its continuous limit. Where $G \neq 0$, the time-local population rate is $\gamma(t) = -2\,\mathrm{Re}(\dot{G}/G)$. Singular rates at amplitude zeros require a regularized or direct state-based definition.

Information backflow is generally defined through increases of distinguishability of suitable state pairs, not merely any population revival [9]. For the specified amplitude-damping family, a qualified negative-rate or state-pair construction can define a future functional. Its population-based proxy must be labeled as such if it does not estimate the full distinguishability criterion. Memory clock λq and detuning ratios can align context; the target can be $\int_t^{t+q} \max(-\gamma(s), 0)ds$ when that integral is well-defined and appropriate. This is distinct from the chance of observing a particular qubit outcome.

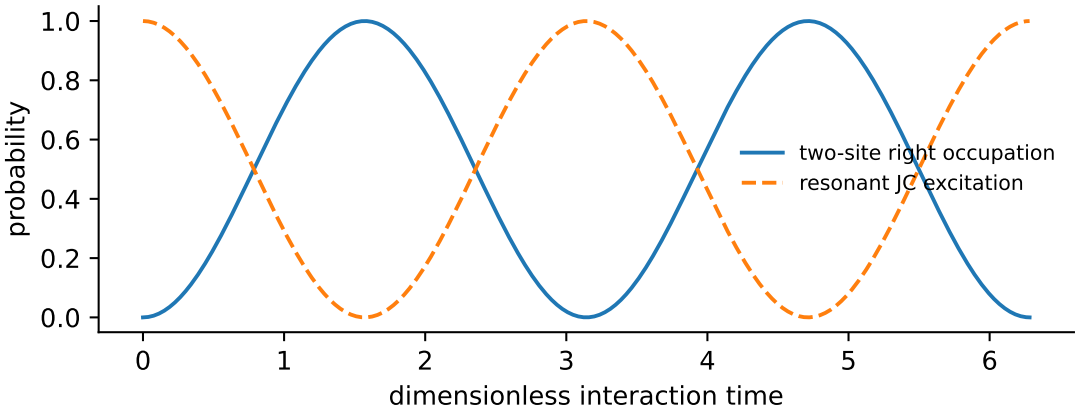


Figure 1: Two exact limiting-model probability functionals at the same dimensionless interaction time. Similar curves do not identify the same physical system. The equations generate this analytical illustration, not the data-driven forecasts.

14 Family-by-family analytical worksheets

The following examples use stipulated states or exact limiting models to work out target arithmetic. They are not predictions obtained from a fitted WPM model. Displaying their physical equations does not feed those equations to the data-driven forecasting branch.

Family	Stipulated example	Target calculation
Free packet	$M = \hbar = s_0 = p_0 = 1$, $x_0 = 0$, $t = 2$	Mean 2; variance $1 + 2^2/4 = 2$
Harmonic oscillator	$M = \hbar = \omega = x_0 = 1$, $p_0 = 0$, coherent state, $t = \pi/2$	Mean 0; variance $1/2$
Barrier scattering	Discrete future mass 0.6 left, 0.4 beyond a boundary	Instantaneous right occupation 0.4; not an asymptotic flux statement
Double well	Two-state hopping with $Jt/\hbar = \pi/6$, initially left	Right occupation $\sin^2(\pi/6) = 1/4$
Multi-well	Masses $(0.2, 0.3, 0.5)$ at $(-1, 0, 1)$	Mean 0.3; variance $0.7 - 0.09 = 0.61$
Interference	Local contributions $a = b = 1/\sqrt{2}$ with relative phase φ	Local intensity $1 + \cos\varphi$; height is not integrated probability
GPE	Normalized initial condensate with zero net boundary flux and real coefficients	Integrated mass remains 1 by the continuity identity
Lindblad	Initially excited qubit, amplitude-damping rate 1, $t = \log 2$	Excitation $1/2$; diagonal purity $1/2$
Ising limiting spin	One spin, $H = -\sigma_x$, initially up, $t = \pi/6$, $\hbar = 1$	$M_z = \cos(2t) = 1/2$; declared boundary equality decides label
Anderson density	Site masses $(1/2, 1/2)$ at $(0, 1)$	IPR $1/2$; SD $1/2$. Mass $(1, 0)$ gives IPR 1 and SD 0
Bose–Hubbard	One particle on two sites, equal onsite terms, $Jt/\hbar = \pi/4$	Right mass $1/2$; onsite two-particle term vanishes in this sector
Hard-core/spinless	Same one-particle two-site sector with equal hopping	Right mass $1/2$; this does not establish many-particle equivalence
Jaynes–Cummings	Resonant $\|e, 0\rangle$, $gt = \pi/4$	Excitation $\cos^2(\pi/4) = 1/2$
Full Rabi limiting case	$g = 0$, initially excited atom	Excitation stays 1; phase evolution alone is not an excitation excursion
Floquet spin	One-period unitary $e^{-i(\pi/4)\sigma_x}$, initially up	After one period excitation $1/2$; after two, 0
Kicked-rotor transport	Stipulated variance increase 4 over two kicks, $K = 2$, $D_{\mathrm{ref}} = K^2/2$	Normalized increment $4/(2 \times 2) = 1$
Backflow limit	Monotone amplitude-damping path $P_e = e^{-t}$	Rate 1; negative-rate backflow functional 0

For actual use, the future values in this table are unknown. The forecast must estimate them from the permitted history, then compute the declared functional while propagating uncertainty. A method that uses the stipulated future state as a feature would not be performing the observation-limited task.

15 Physical constraints and scoring

For a discrete density forecast $\widehat{p}$, project onto $\{p \geq 0, \sum p = 1\}$ if that constraint is required. The Euclidean projection has $p_i = \max(\widehat{p}_i - \tau, 0)$ with τ chosen to make the sum one. KKT stationarity and complementary slackness give this form. On a weighted grid apply the constraint to probability masses or use the corresponding weighted projection; density heights do not sum to one without cell volumes.

Score full densities by total variation or an appropriate proper distribution score; score scalar expectations by an appropriate error; score binary structural events by Brier or log loss. For binary Y with true conditional probability p_*,

$$\mathbb{E}[(Y - p)^2 \mid \mathcal{F}_t] = p_*(1 - p_*) + (p - p_*)^2. \tag{15.1}$$

A density with low TV error can nevertheless miss a finely thresholded event if the true functional lies close to its boundary. Report that sensitivity rather than converting all target errors into one number.

16 Data-driven versus equation-fed prediction: an exact comparison

The optimal conditional probability under a correct fully specified information model minimizes expected Brier score. A data-driven forecast cannot have lower population Brier than that same-information oracle. It can have lower Brier than a misspecified, partially observed, numerically approximate or miscalibrated equation-fed predictor.

Worked example 16.1 (When a data-driven model has lower Brier)**.** Suppose the true event probability is 0.6. An equation-fed approximation gives 0.8, while a data-driven estimate gives 0.65. Their expected Brier scores are $0.24 + 0.2^2 = 0.28$ and $0.24 + 0.05^2 = 0.2425$. The gain follows from smaller probability error, not a violation of quantum dynamics. The correct oracle 0.6 has score 0.24.

Evidence statement. Archived WPM quantum reports describe lower Brier scores in selected tasks against Fixed and observation-fitted conventional baselines [4, 5]. Those statements do not by themselves establish a comparison against a generator-informed quantum propagator; VAR, Ridge and Fixed are not synonymous with equation-fed. Consequently the numerical example above demonstrates a mathematical possibility, and no general empirical superiority to correctly specified quantum equations is claimed.

17 Application procedure and worked questions

1. Fix the measured channels, preparation protocol, target, physical endpoint and loss. Distinguish Born occupation from uncertainty about future occupation.
2. Declare whether known settings are inputs; exclude unobserved future states and the true generator from the data-driven branch.
3. Normalize system coordinates and target semantics separately. Keep independent observables and regime layers intact.
4. Fit target-specific predictors and build landscapes from matured issued errors. Preserve individual flow curves, positions and existence.

5. Use Fixed, Multi or a declared Auto policy within eligible same-target predictions. Enforce physical constraints without interpreting projection as accuracy.

6. Return a forecast with its observation contract, model uncertainty, applicability status and horizon. Diagnose phase ambiguity, sampling aliases and missing channels before increasing model complexity.

Exercise 17.1. For a two-site hopping example with $Jt/\hbar = \pi/4$, find the right-site probability. Does it predict the next individual outcome?

Solution. It is $\sin^2(\pi/4) = 1/2$. An individual outcome remains random under the measurement protocol.

Exercise 17.2. Two site-probability vectors are $(1, 0, 0)$ and $(0, 0, 1)$. Compare IPR and location.

Solution. Both have IPR 1 but distinct centroids. Concentration alone is not a complete spatial state.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study and its quantum companion (Papers 7 and 8 of the series) share the archival DOI 10.5281/zenodo.23783870.

References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction.* Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation.* Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[4] T. Nguyen. *The Multi-Lens W-P-m Framework: A Resolution-Spectrum Extension for Structural Forecasting Across Chaotic and Quantum Dynamics.* Archive version 24: dual normalization and non-Markovian extension. October 2026. Referenced for method provenance and scoped baseline statements, not as a proof of universal superiority.

[5] T. Nguyen. *Target-Conditioned Temporal Resolution in Quantum Forecasting: Location, Scale, Structural Context, and Foreseeable Boundary Risk.* Version 2, October 2026. Archived manuscript.

[6] QuTiP developers. *Using tensor products and partial traces.* Official documentation. `https://qutip.org/docs/4.6/guide/guide-tensor.html`.

[7] QuTiP developers. *Floquet formalism.* Official documentation. `https://qutip.org/docs/4.6/guide/dynamics/dynamics-floquet.html`.

[8] P. W. Anderson. Absence of diffusion in certain random lattices. *Physical Review* 109, 1492–1505, 1958. doi:10.1103/PhysRev.109.1492.

[9] H.-P. Breuer, E.-M. Laine and J. Piilo. Measure for the degree of non-Markovian behavior of quantum processes in open systems. *Physical Review Letters* 103, 210401, 2009. doi:10.1103/PhysRevLett.103.210401.